# Project Objective

To develop an intelligent, explainable AI system that identifies skill gaps from resume analysis, benchmarks against current market demands, and generates personalized learning roadmaps with course recommendations. This bridges academic-industry gaps by highlighting critical market-valued skills for students and job seekers.

### Automated Resume Analysis

Extract skills via NLP from resumes.

### Market-Aware Gap Scoring

Prioritize skill gaps by job market demand.

### Explainable Recommendations

Provide transparent reasoning for suggestions.

### Personalized Roadmaps

Tailored week-by-week learning plans.

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# Problem Statement

Final-year computer science students and recent graduates face a critical challenge: identifying essential technical skills for the rapidly evolving job market due to unspecific and unpersonalized traditional guidance.

### Skill-Job Market Mismatch

Students lack awareness of specific technical skills employers seek, leading to hindered job applications.

### Information Overload

Students struggle to identify effective courses among numerous online learning resources.

### Lack of Personalization

Generic advice ignores individual backgrounds, leading to inefficient learning paths.

### No Actionable Insights

Existing tools list missing skills but lack rationale or structured learning roadmaps.

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# Abstract

This project introduces an explainable AI system that analyzes resumes, identifies critical skill gaps against current job market demands, and generates personalized learning roadmaps with targeted course recommendations. It leverages Natural Language Processing (NLP) and TF-IDF to achieve this, providing transparent reasoning for its recommendations.

AI-Resume-Analyzer

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# Existing System Analysis

Current career guidance and skill gap identification suffer from significant limitations, failing to adapt to rapid technological evolution and market demands.

### Manual Analysis

Manual analysis is slow.

### Generic Platforms

Generic platforms lack personalization.

### Basic Keyword Matching

Basic keyword matching ignores market demand.

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# Proposed System Architecture

Our system is a multi-layered architecture integrating advanced NLP, machine learning, and explainable AI to address current limitations.

1. User Interface
2. Resume Processing
3. Skill Analysis
4. Recommendation
5. Learning Roadmap

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# Literature Review

Foundational research in skill gap analysis and personalized learning informs our approach, highlighting innovation opportunities in explainability and market-awareness.

### Resume Parsing Techniques

Transformer-based NLP models are highly effective in skill extraction.

### Skill Taxonomy Mapping

Comprehensive skill ontologies are crucial for intelligent recommendation systems.

### Explainable AI Systems

Transparency in recommendations significantly boosts user trust.

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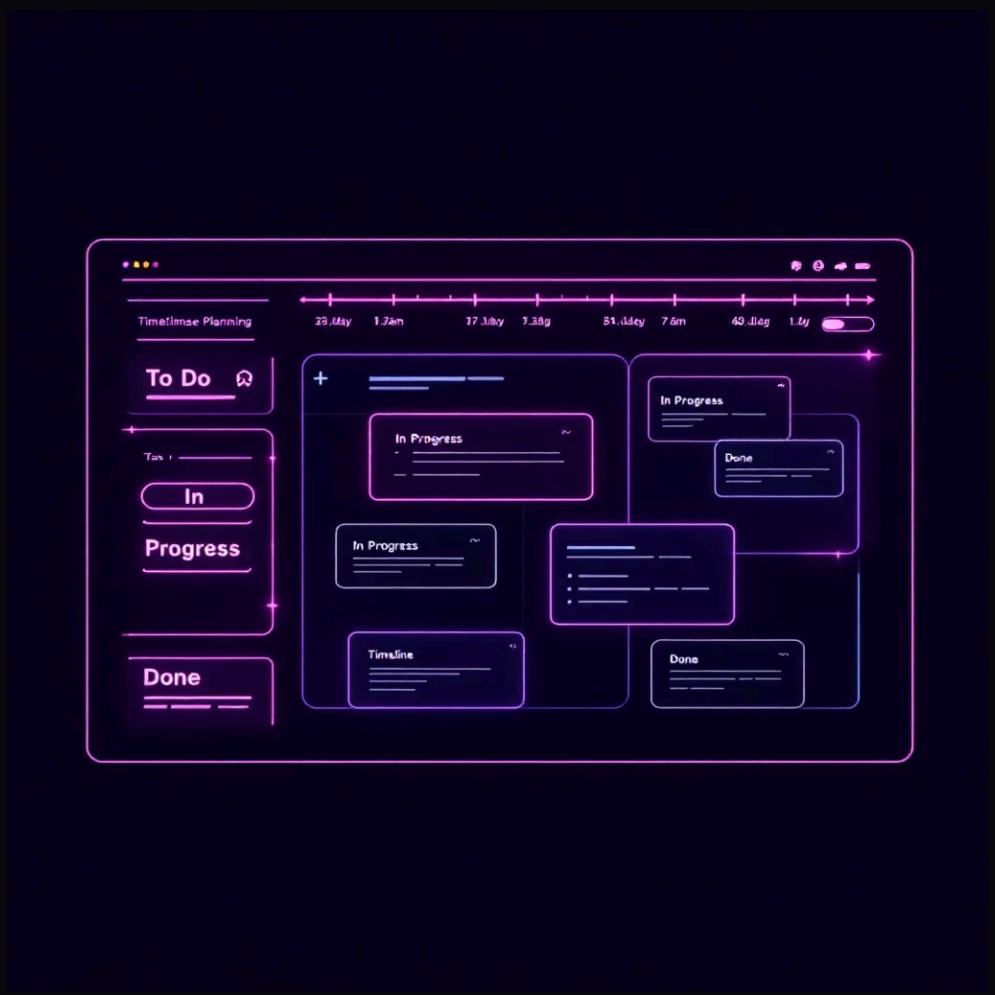
# System Implementation Plan




The project uses an agile methodology with clearly defined six-week milestones, ensuring systematic progress from core functionality to advanced features and deployment. Each sprint delivers a complete, testable module.

1

### Phase 1: Resume Parsing

Setup frontend/backend and implement resume parsing (PyMuPDF, spaCy).

2

### Phase 2: Skill Analysis & Recommendations

Build skill comparison (TF-IDF), basic skill gap identification, market-aware weighting, explainable AI, and course database. Develop personalized learning roadmaps, resume confidence, and interactive dashboards.

3

### Phase 3: Roadmap & Deployment

Conduct comprehensive testing, optimize performance, and deploy the system.

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# Expected Outcomes & Impact

This AI-driven skill gap analysis system enhances career readiness and strategic skill development for students, educators, and the tech workforce.

## Improved Employability

Students gain clear insight into required technical skills, reducing time-to-employment by focusing on high-demand competencies.

## Time Efficiency

Automated analysis provides instant, data-driven insights, replacing manual research and freeing students to focus on skill development.

## Data-Driven Decisions

Market analytics-backed recommendations enable informed decisions on course investments, prioritizing skills for maximum career impact.

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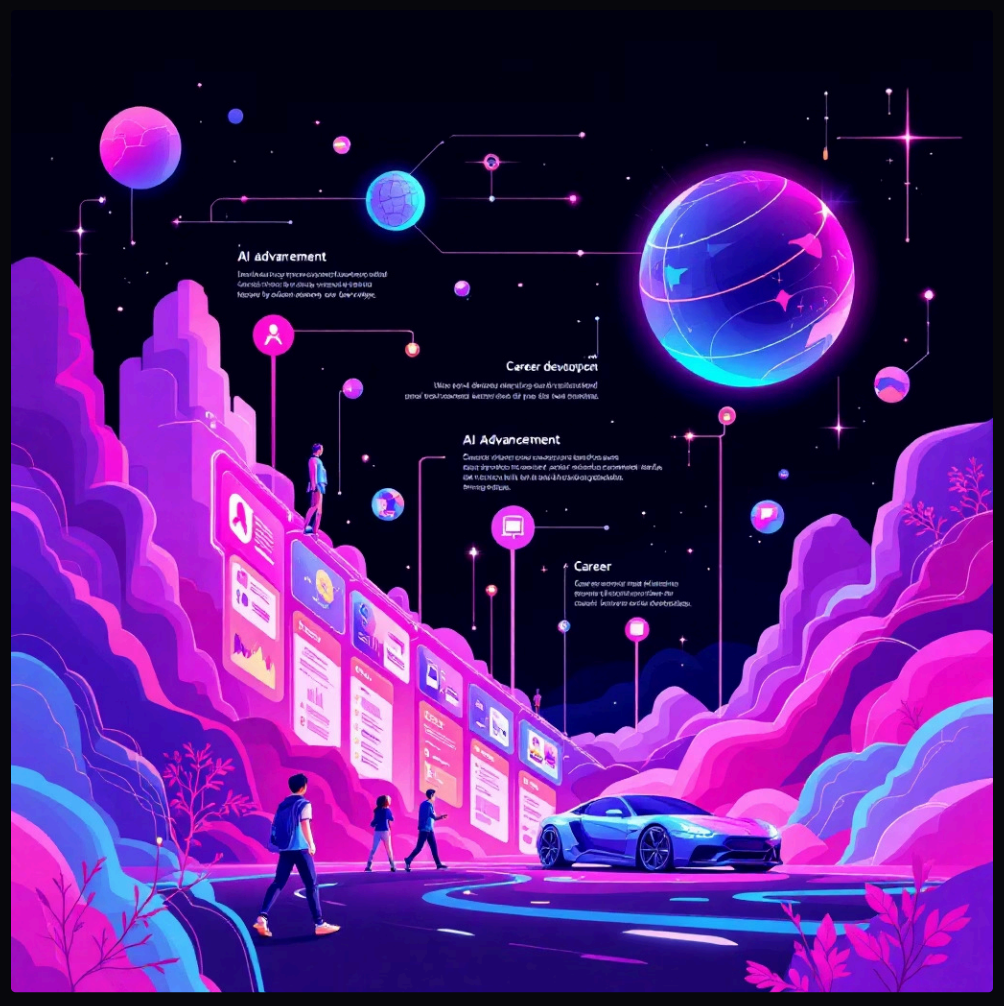
# Conclusion & Future Scope

This project demonstrates how artificial intelligence, natural language processing, and explainable recommendation systems provide a practical solution for the critical tech skill gap faced by computer science students. Our system offers intelligent, market-aware career guidance, innovating beyond existing solutions with personalized learning pathways and resume confidence analysis.




### Current Achievement

Functional AI system with NLP resume analysis, market-aware gap scoring, and explainable recommendations successfully deployed.

### Future Enhancements

Integration of real-time job scraping, LLM-based personalization, career path simulation, and predictive salary impact analysis.

### Scale & Impact

Platform expansion to multiple languages, industries beyond tech, and partnership with educational institutions for widespread adoption.

## Future Development Roadmap

01

### LLM Integration

Incorporate GPT-4 or Gemini for advanced natural language explanations and conversational guidance.

02

### Real-Time Scraping

Automated job posting aggregation from platforms like LinkedIn and Indeed for continuously updated market data.

03

### Salary Prediction

ML models to estimate potential salary improvements based on skill acquisition.

04

### Mobile Application

Develop React Native mobile apps for iOS and Android to improve accessibility and user engagement.

**Final Thoughts:** This project exemplifies how computer science students can leverage AI to solve real-world problems affecting millions of graduates, demonstrating mastery of NLP, machine learning, and full-stack development.

> *"The best way to predict your future is to create it through strategic skill development guided by data-driven insights."*

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